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● *Clinical Original Contribution*

TREATMENT OF CHOROIDAL MELANOMA WITH I-125 PLAQUE

JAMES FONTANESI, M.D.,[1] DAVID MEYER, M.D.,[2] SHIZHAO XU, M.D.[2] AND DOUG TAI, PH.D.[1]

[1]Department of Radiation Oncology, University of Tennessee College of Medicine; and [2]Vitreoretinal Foundation, Memphis, TN

Purposes: To evaluate efficacy of I-125 episcleral plaque therapy in patients with ocular melanoma and determine survival, eventual visual acuity, and complications.
Methods and Materials: Between July 1, 1984 and January 1, 1991, 144 patients with diagnosis of ocular melanoma were treated with high activity I-125 episcleral plaques. Tumor volumes ranged from 14 to 3449 mm^3. Lesion size included small (n = 15; height < 5 mm, and/or largest basal diameter of 8–16 mm) and large (n = 45; height > 8 mm, and/or largest basal diameter > 16 mm). Apical doses ranged from 74.25 to 83.66 Gy with scleral doses ranging from 41 and 160 Gy. Follow-up has ranged from 25 to 90 months (Med = 46 months).
Results: Ocular survival was noted in 130/144. Reasons for enucleation included progressive tumor growth (n = 4), extrascleral extension (n = 4), or blind/painful eye (n = 6), 94 patients developed complications which included cataract (n = 43), optic neuropathy (n = 12), neovascular glaucoma (n = 8) and retinopathies (n = 31). Visual acuity testing pre-episcleral plaque therapy revealed 102 patients with 20/200 vision; at last follow-up 59 patients demonstrated visual acuity testing of 20/200 or better.
Conclusion: The use of episcleral I-125 plaque therapy allows for safe and effective therapy in patients with ocular melanoma of various size depending on location and probable visual acuity outcome. A total apical dose of 75 Gy given at 60–65 cGy/hour provides durable local control with acceptable complication rates.

Iodine-125, Brachytherapy, Melanoma, Ocular.

INTRODUCTION

Although episcleral plaque therapy (EPT) has been effectively used for over 20 years (10, 11, 13, 17) there still persists controversy as to when EPT is best used. A present ongoing study (Collaborative Ocular Melanoma Study) which includes 20 clinical centers evaluating EPT versus enucleation for patients with median and preoperative irradiation for large lesions is presently undergoing evaluation (7). Various investigators have reported results with various plaques and use of different isotopes for various tumor sizes (2, 5). There have also been reports which relate tumor size, total apical dose, and location of lesion (i.e., proximity to fovea, optic nerve, macula) with eventual local control, visual acuity, and complications (1, 3, 14). There has also been recent introduction of microwave induced hypothermia therapy (MIHT) to supplement the effect of EPT (8). We present a series of 144 patients treated with high activity I-125 with varying hourly and total dose rates to the tumor apices and report on the subsequent local control, visual acuity and complications.

METHODS AND MATERIALS

Between July 1, 1984 and January 1, 1990, 144 patients with diagnosis of ocular melanoma have been referred to the same ophthalmologist (DM) for evaluation. Each patient underwent detailed history and physical examination. The clinical diagnosis of malignant melanoma was made by documentation of tumor growth patterns, ophthalmoscopic evaluation, fluorescein angiography, funduscopic photography, and quantitative A-scan ultrasonography. In addition, visual acuity testing, intraocular pressures, and direct visualization which allowed for detailed identification of location and proximity to important sites, size of basal component, height of tumor, shape of tumor, presence of rupture of Bruch's membrane degree of retinal detachment or effusion if present were made. Following this evaluation each patient was informed of treatment options. If conservative therapy with high activity I-125 was chosen, the patient underwent metastatic workup including liver enzyme profile, chest X ray, CT scan of the head, abdomen and pelvis. If any irregularities

Reprint requests to: James Fontanesi, M.D., Department of Radiation Oncology, St. Jude Children's Research Hospital, 332 N. Lauderdale, Memphis, TN 38101.

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